

GCF Factor Grid Challenge

Unit 1: Number System Launch • Lesson 1-2 • EduWonderLab

UNIT 1 • LESSON 1-2 • TEACHER NOTES

Standard 6.NS.4

FORMAT Partner capture game	TIME 25–30 min
GROUPING Pairs	TOPIC Greatest Common Factor
STANDARD 6.NS.4	PREP LEVEL Light — print number-pair cards and factor grids

Learning goal *I can find the greatest common factor of two numbers and explain why it is greatest.*

Teacher setup

- Print one card set, two factor grids, and highlighters per pair. Shuffle the cards face down.
- Players take turns flipping a card and listing the factors of both numbers on the grid.
- A card is only “captured” after the player names the GCF and explains why it is the greatest.

Materials

Number-pair cards, Factor grids, Highlighters, Pencil.

Differentiation & language support

- Provide a factor rainbow example so students list factors in pairs (SPED).
- Sentence stem: “The GCF is ____ because it is the largest factor both numbers share.”
- ESOL: highlight common factors in one color, the greatest in a second color.

Key vocabulary (English / Español):

Term	Español	What it means
factor	factor	a number that divides evenly into another
common factor	factor común	a factor shared by two numbers
greatest common factor	máximo común divisor	the largest shared factor

Quick teacher check: *Ask one pair to prove their largest factor really divides both numbers.*

GCF Factor Grid Challenge

Unit 1: Number System Launch · Lesson 1-2 · EduWonderLab

Name: _____

Date: _____

Class: _____

Your goal: *I can find the greatest common factor of two numbers and explain why it is greatest.*

What you need

Number-pair cards, Factor grid, Highlighter.

How to play

1. Flip a card. List all factors of each number on your grid.
2. Highlight the factors that appear on both lists.
3. Circle the largest highlighted factor — that is the GCF.
4. Say the GCF out loud and explain why it is greatest to capture the card.

Number-Pair Cards

List the factors of both numbers, circle the ones they share, then capture the card by naming the GCF.

1. 12 and 18	2. 24 and 36
3. 16 and 40	4. 30 and 45
5. 14 and 35	6. 27 and 36
7. 20 and 50	8. 48 and 60
9. 21 and 28	10. 15 and 50

Record your work (factors of each number):

Explain your thinking

Choose one card. Explain how you know your GCF is the greatest, not just any common factor.

Sentence frame: *The GCF of ___ and ___ is ___ because it is the largest factor they both share.*

★ **Challenge:** Find two numbers whose GCF is 1. What do we call numbers like these? (relatively prime)

ANSWER KEY

Teacher copy — please do not distribute to students.

UNIT 1 • LESSON 1-2 • GCF Factor Grid Challenge

Standard 6.NS.4

Answers

1. 12 and 18 Answer: 6	2. 24 and 36 Answer: 12
3. 16 and 40 Answer: 8	4. 30 and 45 Answer: 15
5. 14 and 35 Answer: 7	6. 27 and 36 Answer: 9
7. 20 and 50 Answer: 10	8. 48 and 60 Answer: 12
9. 21 and 28 Answer: 7	10. 15 and 50 Answer: 5

Teaching notes & look-fors

- Students sometimes pick a common factor that is not the largest — require the full factor list.
- Challenge answer: numbers with GCF 1 are relatively prime (e.g., 8 and 15).